

GRAFTON RESEARCH STATION, Australia

WMO: 955710

Lat: 29.6225S

Lon: 152.9606E

Elev: 26

StdP: 101.02

Time Zone: 10.00 (AUS)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 7             | 5.1        | 6.4 | -2.5                          | 3.1 | 15.7 | -0.7 | 3.6 | 14.7 | 9.7                   | 16.6 | 8.4 | 16.9 | 2.0                   | 290  | 0.344 |

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 1             | 9.6                    | 34.6            | 22.8 | 32.5 | 22.6 | 30.8 | 22.2 | 24.8                | 31.6 | 24.1 | 30.1 | 23.3 | 28.6 | 4.3                  | 70   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 23.0                            | 17.8 | 27.3 | 22.3 | 17.1 | 26.8 | 21.7 | 16.4 | 26.1 | 75.4          | 31.8 | 72.4 | 29.9 | 69.5 | 28.7 | 28.0              |

Extreme Annual Design Conditions

| Extreme Annual WS |      |                    | Extreme Annual Temperature |           |      |            | n-Year Return Period Values of Extreme Temperature |            |      |            |      |      |      |      |      |  |
|-------------------|------|--------------------|----------------------------|-----------|------|------------|----------------------------------------------------|------------|------|------------|------|------|------|------|------|--|
| Mean              |      | Standard Deviation |                            | n=5 years |      | n=10 years |                                                    | n=20 years |      | n=50 years |      |      |      |      |      |  |
| 1%                | 2.5% | 5%                 | Min                        | Max       | Min  | Max        | Min                                                | Max        | Min  | Max        | Min  | Max  |      |      |      |  |
| (a)               | (b)  | (c)                | (e)                        | (f)       | (g)  | (h)        | (i)                                                | (j)        | (k)  | (l)        | (m)  | (n)  | (o)  | (p)  |      |  |
| 8.7               | 7.4  | 6.4                | DB                         | 2.0       | 39.4 | 1.1        | 2.5                                                | 1.2        | 41.1 | 0.6        | 42.6 | 0.0  | 44.0 | -0.8 | 45.7 |  |
|                   |      |                    | WB                         | 1.1       | 26.2 | 1.2        | 0.9                                                | 0.2        | 26.9 | -0.4       | 27.4 | -1.1 | 27.9 | -1.9 | 28.5 |  |

Monthly Climatic Design Conditions

|                                                                   |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|-------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                   |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 19.7   | 24.8  | 24.4  | 22.9  | 20.2  | 17.0  | 14.8  | 14.1  | 15.1  | 17.9  | 20.1  | 22.2  | 23.6 |
|                                                                   | DBStd       | 4.34   | 2.28  | 2.17  | 1.90  | 1.84  | 1.96  | 1.95  | 1.91  | 2.06  | 2.33  | 2.61  | 2.57  | 2.51 |
|                                                                   | HDD10.0     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                   | HDD18.3     | 448    | 0     | 0     | 0     | 4     | 52    | 108   | 132   | 104   | 34    | 11    | 2     | 1    |
|                                                                   | CDD10.0     | 3551   | 458   | 402   | 401   | 306   | 216   | 143   | 127   | 157   | 238   | 314   | 367   | 423  |
|                                                                   | CDD18.3     | 957    | 199   | 169   | 142   | 61    | 9     | 1     | 1     | 3     | 23    | 67    | 118   | 165  |
|                                                                   | CDH23.3     | 6578   | 1503  | 1172  | 761   | 262   | 57    | 3     | 7     | 62    | 232   | 520   | 833   | 1164 |
|                                                                   | CDH26.7     | 2187   | 578   | 404   | 195   | 34    | 4     | 1     | 0     | 11    | 61    | 173   | 299   | 428  |
| Wind                                                              | WSAvg       | 2.7    | 2.7   | 2.9   | 2.6   | 2.6   | 2.4   | 2.7   | 2.6   | 2.8   | 2.8   | 2.8   | 2.8   | 2.8  |
| Precipitation                                                     | PrecAvg     | 1071   | 130   | 124   | 132   | 115   | 91    | 71    | 56    | 37    | 31    | 75    | 89    | 106  |
|                                                                   | PrecMax     | 1695   | 303   | 359   | 518   | 408   | 352   | 589   | 295   | 160   | 99    | 573   | 330   | 348  |
|                                                                   | PrecMin     | 495    | 8     | 16    | 2     | 10    | 13    | 4     | 0     | 0     | 0     | 4     | 10    | 24   |
|                                                                   | PrecStd     | 280    | 78    | 94    | 98    | 113   | 90    | 110   | 69    | 40    | 29    | 100   | 67    | 72   |
| Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 37.6  | 36.2  | 33.5  | 29.4  | 27.1  | 23.4  | 24.4  | 28.0  | 31.3  | 34.1  | 36.0  | 37.8 |
|                                                                   |             | MCWB   | 24.1  | 24.7  | 23.5  | 20.9  | 18.4  | 16.0  | 15.5  | 15.5  | 17.8  | 20.0  | 22.0  | 22.3 |
|                                                                   | 2%          | DB     | 33.7  | 32.9  | 30.4  | 27.5  | 24.9  | 21.8  | 22.3  | 24.9  | 28.1  | 30.9  | 32.4  | 33.3 |
|                                                                   |             | MCWB   | 23.9  | 24.2  | 22.7  | 20.0  | 17.4  | 15.4  | 14.3  | 14.8  | 17.0  | 19.3  | 21.2  | 22.8 |
|                                                                   | 5%          | DB     | 31.5  | 30.8  | 28.6  | 26.0  | 23.3  | 20.7  | 20.6  | 22.8  | 25.8  | 28.3  | 29.9  | 30.6 |
|                                                                   |             | MCWB   | 23.4  | 22.9  | 21.8  | 19.3  | 16.6  | 14.7  | 13.7  | 14.0  | 16.6  | 18.8  | 20.9  | 22.1 |
|                                                                   | 10%         | DB     | 29.8  | 29.1  | 27.2  | 24.7  | 22.0  | 19.5  | 19.2  | 21.1  | 23.9  | 26.1  | 27.7  | 28.8 |
|                                                                   |             | MCWB   | 22.6  | 22.0  | 21.0  | 18.7  | 15.8  | 14.2  | 13.1  | 13.4  | 16.1  | 18.4  | 20.2  | 21.5 |
| Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 25.9  | 25.9  | 24.6  | 22.2  | 20.2  | 18.1  | 18.2  | 18.0  | 20.1  | 22.4  | 24.3  | 25.3 |
|                                                                   |             | MCDB   | 33.2  | 33.3  | 30.5  | 27.4  | 24.2  | 20.8  | 21.5  | 23.7  | 27.0  | 29.1  | 32.7  | 32.7 |
|                                                                   | 2%          | WB     | 24.8  | 24.7  | 23.6  | 21.1  | 19.0  | 17.0  | 16.5  | 16.3  | 18.9  | 21.0  | 22.9  | 24.1 |
|                                                                   |             | MCDB   | 31.5  | 31.1  | 28.4  | 25.4  | 22.7  | 20.0  | 19.5  | 21.7  | 25.3  | 27.4  | 29.5  | 30.8 |
|                                                                   | 5%          | WB     | 24.0  | 23.8  | 22.9  | 20.2  | 17.9  | 16.1  | 15.2  | 15.3  | 18.0  | 20.0  | 21.8  | 23.1 |
|                                                                   |             | MCDB   | 29.7  | 29.1  | 26.9  | 24.4  | 21.4  | 18.9  | 18.7  | 20.2  | 23.6  | 25.8  | 27.2  | 28.8 |
|                                                                   | 10%         | WB     | 23.2  | 22.9  | 22.0  | 19.5  | 17.0  | 15.4  | 14.1  | 14.5  | 17.0  | 19.1  | 21.0  | 22.3 |
|                                                                   |             | MCDB   | 28.2  | 27.7  | 25.9  | 23.5  | 20.5  | 18.1  | 17.7  | 19.2  | 21.8  | 24.3  | 26.0  | 27.1 |
| Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 9.6   | 9.4   | 9.0   | 9.7   | 10.9  | 10.3  | 11.8  | 13.2  | 12.6  | 11.6  | 11.0  | 10.1 |
|                                                                   |             | MCDBR  | 13.1  | 12.4  | 11.5  | 11.5  | 12.6  | 11.9  | 13.8  | 16.3  | 16.3  | 15.9  | 15.0  | 13.8 |
|                                                                   | 5% WB       | MCWBR  | 4.9   | 4.5   | 4.5   | 5.1   | 6.1   | 6.5   | 7.1   | 7.6   | 6.9   | 6.4   | 5.7   | 5.1  |
|                                                                   |             | MCWBR  | 11.5  | 11.5  | 10.3  | 10.1  | 10.3  | 9.3   | 10.9  | 13.2  | 13.6  | 13.0  | 12.4  | 11.8 |
| Clear-Sky Solar Irradiance                                        | taub        | 0.386  | 0.373 | 0.356 | 0.336 | 0.310 | 0.304 | 0.297 | 0.311 | 0.355 | 0.388 | 0.385 | 0.389 |      |
|                                                                   |             | 2.505  | 2.559 | 2.599 | 2.627 | 2.654 | 2.660 | 2.662 | 2.601 | 2.471 | 2.413 | 2.473 | 2.484 |      |
|                                                                   | Ebn at Noon | 955    | 950   | 935   | 906   | 886   | 870   | 893   | 919   | 920   | 925   | 950   | 955   |      |
|                                                                   |             | 114    | 105   | 96    | 84    | 74    | 69    | 72    | 84    | 106   | 120   | 117   | 117   |      |
| All-Sky Solar Radiation                                           | RadAvg      | 6.49   | 5.88  | 4.97  | 4.26  | 3.49  | 2.91  | 3.34  | 4.25  | 5.31  | 5.90  | 6.32  | 6.48  |      |
|                                                                   | RadStd      | 0.54   | 0.48  | 0.31  | 0.21  | 0.23  | 0.18  | 0.26  | 0.31  | 0.39  | 0.54  | 0.60  | 0.61  |      |

Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| (46) Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (17 neighbors) | +0.35 | N/A     | N/A    | +0.63   | N/A   | N/A   | N/A         | -48     | +161    | +94     |

Nomenclature: See separate page